OrcSDR Tab5 - Bench Test Plan

Covers everything changed on **2026-11-03**: the fork re-based onto upstream **cdc8181** (v0.5.0-beta.1 plus 95 commits, driver 0.9.4) with this fork's work re-applied. Branch **main** at commit **6e03b90d**. It **has** since been compiled, flashed and run: a clean build gives 0 errors with about 30% of the app partition free, and the device boots with **71 of 71 self-checks passing** and SD, Wi-Fi and the RTL-SDR V4 all up. Sections 4, 7 and much of 6, 8 and 9 have been exercised over the serial link; what remains below needs a human - a touchscreen, a transmitter, a second network, or a particular day of the week.

Before flashing: back up the firmware currently on the Tab5 (the flash skill's backup step, or `esptool read_flash`), so you can return to it. If the new build misbehaves badly, the last pre-consolidation line is commit **a7e2db6** - still in history, buildable with `git checkout a7e2db6`. Keep a serial monitor open for every test and save the log; most results below are serial lines. Pair the serial session first (`tools/tab5_serial_auth.ps1`) so commands marked *(auth)* are accepted.

Reading the sheet. Rows on a green background were already run over the serial link on 2026-11-03 and come filled in, with the evidence in Notes. **PASS** means it matched the Expected column. **PART** means the serial half passed and something visual or physical is still open - the Notes say which. **N/A** means there was nothing to test. Every row still showing [ ] / [ ] needs a person at the bench: a touchscreen, a transmitter, a second router, or a particular day of the week.

1. Build and size

The one gate that decides whether this branch can ship.

#	Test	How	Expected	P / F	Notes
1	Clean build	Build with your usual script on the Windows PC.	0 errors. Note the warning count; new ones in `channel_scan_plan.cpp`, `main.cpp` scanner code or `home_dashboard.cpp` are mine to fix.	[ ] / [ ]	
2	App partition headroom	Read the size report at the end of the build.	Record % free. The last measured build had **33%**; upstream added 17 always-built files. Under ~10% is a problem.	[ ] / [ ]	
3	Host tests	`bash tools/test-band-profile.sh` (WSL or Git Bash).	`channel_scan_plan PASS` plus band_profile and gmrs_channels PASS.	[ ] / [ ]	

2. Boot

Watch the serial log from reset to the Home screen.

#	Test	How	Expected	P / F	Notes
1	Boot reaches Home	Power-cycle with the RTL-SDR attached.	Splash, then Home with a live spectrum. `BOOT_SPLASH_SUMMARY` shows sd/wifi/rtl.	[ ] / [ ]	
2	Self-checks honest	Scan the log for `_SELF_CHECK` lines.	Each check prints **exactly one** line, _OK or _FAIL - never both. Note every _FAIL.	[ ] / [ ]	
3	New self-checks	Look for the two new ones.	`ORC_SAME_SELF_CHECK_OK` and `ORC_CHANNEL_SCAN_SELF_CHECK_OK`.	[ ] / [ ]	
4	A failing check no longer stops boot	Only if one of the shortwave/CB/airband/receiver-controls checks reports _FAIL.	Boot still finishes: Wi-Fi, SD and radio all come up anyway.	[ ] / [ ]	
5	Recent dashboards	Open the dashboard list after upgrading.	A GMRS entry may show as AM and AM as FT8 once (known, cosmetic). Open a few; it settles.	[ ] / [ ]	

3. Wi-Fi

#	Test	How	Expected	P / F	Notes
1	Connect to your router	Settings - Connectivity - connect to your saved network.	Connects (about 1 s slower than before: it scans all channels and takes the strongest AP).	**PASS**	`connected=1` with `auto_connect=1`; rejoined unaided across three reboots.
2	WPA-only router (if you have one)	Join an older WPA (not WPA2) network.	Joins. Before, ESP-IDF's WPA2-only default would refuse it.	[ ] / [ ]	
3	Short password refused	ADD HIDDEN / connect, type a 6-character password, CONNECT.	Keyboard stays open titled `PASSWORD: 8+ CHARACTERS`. 5 characters is allowed (WEP).	[ ] / [ ]	
4	Drop reason shown	With Wi-Fi connected, switch the router off or walk out of range.	Settings shows **Lost: <reason> - reconnecting**; serial `RTL_WIFI_DISCONNECTED reason=N`. Reconnects when the AP is back.	[ ] / [ ]	
5	Wrong password reason	Connect with a wrong password.	**Failed: wrong password** (or handshake timeout), not a bare 'Connection failed'.	[ ] / [ ]	
6	RESET LINK button	Settings - Connectivity - RESET LINK (beside CONNECT ON BOOT).	Shows RESETTING..., serial `RTL_WIFI_RESET_LINK queued` then `RTL_WIFI_LINK_RECOVERY result=ok`; reconnects on its own.	[ ] / [ ]	
7	RESET LINK over serial	`RTL_WIFI_RESET_LINK` (auth).	Same as above.	[ ] / [ ]	
8	Antenna setting persists	Set WI-FI ANTENNA to EXTERNAL (MMCX), reboot.	Still EXTERNAL after reboot. **Previously it always reverted to INTERNAL.**	**PASS**	`antenna=external` before and after reboot, and the restore to `internal` persisted too. Wi-Fi still read `connected=1` with EXTERNAL selected and **no MMCX antenna fitted**, so this setting alone does not prove an external antenna is in use.
9	Long SSID layout	Save or scan a network with a long name.	Status line, saved rows and scan buttons do not overlap their neighbours or the CONNECT button.	[ ] / [ ]	

4. Companion (LAN web console)

Settings - Accessories & Companion. Use a browser and a PC on the same LAN.

#	Test	How	Expected	P / F	Notes
1	Default is view-only	Enable the web console. Check the REMOTE CONTROL row.	Reads **VIEW ONLY**. The page at http://<tab5-ip>/ shows status, spectrum, audio.	**PASS**	Straight after `RTL_WEB ON`: `control=0`.
2	Control refused when off	From the PC: `curl -X POST --data volume_up http://<ip>/api/action`	`403` with body `control disabled`.	**PASS**	Exactly `403` / `control disabled`.
3	Control works when on	Tap ALLOW (or `RTL_WEB CONTROL ON`), repeat the curl.	`202` / ok, and the volume changes on the Tab5.	**PASS**	`202` / `ok`, and volume really moved **128 to 144** (read back from `/api/status`, then restored).
4	Status reports it	`RTL_WEB_STATUS`	Ends with `control=0` or `control=1`.	**PASS**	`control=0` before ALLOW, `control=1` after.
5	Status JSON valid	Open http://<ip>/api/status in the browser for a minute on several bands.	Always parses (no `nan`); page never blanks.	**PASS**	18 of 18 samples parsed across WX and FM, 31 keys, no `nan` or `inf`. Allow a long timeout: the link idles slowly and 12 s was not enough to connect.

5. FM broadcast

Rows 1-6 are touch-only: there is no serial verb that starts a seek, so `RTL_FM_SEEK` appears in the log as a *result* and cannot be used to drive one. They have to be done at the panel.

#	Test	How	Expected	P / F	Notes
1	SEEK up	FM dashboard, Listen tab. Tap SEEK + >> from a station.	Both SEEK buttons read STOP SEEK while it sweeps; stops on the next real station **above**. Serial `RTL_FM_SEEK found frequency_hz=...`	[ ] / [ ]	
2	SEEK down	Tap << SEEK -.	Stops on the next real station below.	[ ] / [ ]	
3	Wraps at the edges	Seek up from 107.9; seek down from 88.1.	Wraps round and finds a real station.	[ ] / [ ]	
4	Stop mid-sweep	Tap either SEEK button while it is sweeping.	Stops where it is; labels return to SEEK.	[ ] / [ ]	
5	Preset scan	Run the preset scan.	Finds your real local stations only - not every channel (the old -70 dBFS test passed noise).	[ ] / [ ]	
6	Long RadioText	Station with long RDS RadioText, Listen tab.	Text stays inside its area; does not spill into STEREO VU.	[ ] / [ ]	

#	Test	How	Expected	P / F	Notes
7	RTL_PRESET_TUNE	`RTL_PRESET_LIST` to see how many you have, then `RTL_PRESET_TUNE 1` and `RTL_PRESET_TUNE <last>` (auth).	`RTL_PRESET_TUNE_OK index=1 frequency_hz=..` and the same for the last index, and the dial really moves there - check `RTL_PRESET_LIST` against the frequency on screen. Before, it never matched. **The store holds at most 10** (`RTL_PRESET_SCAN` collects up to 10 stations), so an index above that is not a failure: `RTL_PRESET_TUNE 12` answers `RTL_PRESET_TUNE_INVALID` by design.	**PASS**	`index=1` gave 89.3 MHz and `index=10` gave 107.7 MHz; a live screenshot confirmed the dial actually sat at **107.700**. `RTL_PRESET_TUNE 12` refused correctly.

6. GMRS / marine channel scanner (new - calibrate first)

Thresholds are **placeholders** (min SNR 20 dB, squelch -75 dBFS) from the old DSP chain. Calibrate them before judging the scanner. You need something that transmits on GMRS/FRS - an FRS handheld on low power a few metres away is ideal (receive-only on the Tab5; transmitting on GMRS channels needs your own licence).

#	Test	How	Expected	P / F	Notes
1	SCAN button appears	Home: tune 462.5625 MHz (FRS 1).	The STEP SIZE box becomes **CH 1 / SCAN**. On other bands it is the normal stepper.	**PART**	Band detection confirmed over serial: `band=GMRS channel=1` after `RTL_TUNE BROWSE 462562500`. The **on-screen box still needs your eyes**.
2	Noise reading	On an idle channel: `RTL_CHANNEL_PROBE` five times.	Note the **highest** snr_db, not the average - that is what a threshold has to clear. Measured on one bench V4 at FRS 1: 13.6 / 14.7 / 15.7 / 16.9 / **23.3** dB, all busy=0. Note that 23.3 is above the 20 dB default, so **noise alone can reach the placeholder threshold**; busy stayed 0 only because the peak sat 486-814 kHz off channel against a `tolerance_hz=6250`. Expect single-digit to low-20s dB and set the threshold from your own worst reading.	**PASS**	Five probes run; numbers opposite are the measurement. **Act on it:** the 20 dB default is below the worst noise reading seen here.
3	Signal reading	Key the handheld on that channel; probe again.	Note snr_db (signal), offset_hz near 0, busy=1 if above the threshold.	[ ] / [ ]	
4	Set the threshold	`RTL_CHANNEL_MIN_SNR <dB>` (auth) halfway between the two readings.	`RTL_CHANNEL_MIN_SNR_OK`. Survives a reboot (check `RTL_CHANNEL_MIN_SNR`).	**PASS**	Command path verified (`RTL_CHANNEL_MIN_SNR_OK db=20.0`, reads back after reboot). **The value itself is still the placeholder** - it needs row 3 first.
5	Scan and stop	Tap SCAN, then key the handheld on another channel.	Shows SCANNING - STOP, the dial walks channels, then **HOLD** on the busy channel; `RTL_CHANNEL_SCAN hold`.	[ ] / [ ]	
6	Resume	Stop transmitting.	About 2.5 s later it resumes: `resume ... reason=quiet`.	[ ] / [ ]	

#	Test	How	Expected	P / F	Notes
7	Stop	Tap the SCAN box while scanning or holding.	Stops; box reads SCAN.	[ ] / [ ]	
8	Tune by hand stops it	While scanning, tap the spectrum or step the dial.	Scan stops.	[ ] / [ ]	
9	Lockout	`RTL_CHANNEL_LOCK R15`, then `RTL_CHANNEL_LOCK_LIST`; reboot.	R15 listed and skipped by scans; still locked after reboot. `RTL_CHANNEL_UNLOCK R15` undoes. The lock commands need a GMRS/marine frequency tuned first, or they answer `RTL_CHANNEL_LOCK_ERROR not_gmrs_or_marine`.	**PASS**	Locked, listed as `locked=1 R15`, **still locked after a reboot**, then unlocked again.
10	Old lockouts kept	If you had GMRS lockouts on the old firmware: `RTL_CHANNEL_LOCK_LIST`.	The same channels are still locked.	**N/A**	No lockouts existed to carry over (`locked=0`).
11	Dwell is long enough	After a few minutes of scanning: `RTL_CHANNEL_SCAN_STATUS`.	`stale_skips` stays near 0. If it climbs, the 400 ms dwell is too short - tell me.	**PART**	`stale_skips=0` at rest. **Only a real scan settles this**, which needs row 5.
12	Marine	Tune 156.800 (ch 16) and repeat SCAN/probe if you have marine traffic or a handheld.	CH 16 / SCAN appears; stops on traffic.	[ ] / [ ]	

7. NOAA weather and SAME alerts (new)

#	Test	How	Expected	P / F	Notes
1	Decoder self-test	`RTL_SAME_SELFTEST`	`RTL_SAME_SELFTEST ok failed_step=none`.	**PASS**	Exact string returned.
2	Weekly test alert	Listen on your local NOAA channel during the Required Weekly Test (usually Wednesday late morning; check your NWS office).	Serial `RTL_SAME alert org=WXR event=RWT ...`; `RTL_SAME_LAST` lists your county FIPS.	[ ] / [ ]	
3	Status	`RTL_SAME_STATUS` while on the weather band.	listening=1; counters move after a burst.	**PART**	`band=WX listening=1` after `RTL_TUNE WX 162550000`. Counters are all 0 until a real burst, so the second half waits on row 2.

8. Radio commands and other bands

#	Test	How	Expected	P / F	Notes
1	LoRa modem	`RTL_LORA_MODEM`, then `RTL_LORA_MODEM 11 250000` (auth).	Status line, then `RTL_LORA_MODEM_OK sf=11 bandwidth_hz=250000`.	**PASS**	Both halves returned their exact strings.
2	LoRa self-test	`RTL_LORA_SELFTEST`	`RTL_LORA_SELFTEST ok`.	**PASS**	`ok elapsed_ms=1`.
3	CB squelch	`RTL_SQUELCH -60` (auth), reboot, `RTL_SQUELCH`.	`RTL_SQUELCH_STATUS dbfs=-60`; CB dashboard squelch matches.	**PART**	`dbfs=-60` survived a reboot, then restored to `-75` and that persisted too. **The CB dashboard half still needs your eyes.**

#	Test	How	Expected	P / F	Notes
4	ADS-B positions	ADS-B with your receiver location set, 15+ minutes.	No aircraft jumps across the radar. At TRACE, `RTL_ADSB_POSITION_REJECTED` is rare.	[ ] / [ ]	
5	Driver 0.9.4 on your V4	`RTL_DRIVER STATUS`; then listen on FM, NOAA, airband.	`installed=1 version=0.9.4` and reception as good as before. Check the `RTL_OFFSET_STATUS` line that follows: on the bench V4 it reads `cap=0` / `result=ESP_RTL_SDR_ERR_UNSUPPORTED`, i.e. **offset tuning is not active on this tuner**, so `RTL_DRIVER OFFSET OFF` changes nothing and is not the explanation for a bad signal. Only where `cap=1` is that fallback worth comparing.	**PART**	`installed=1 version=0.9.4` confirmed, and `cap=0` / `UNSUPPORTED` recorded. **Listening on each band is still yours to judge.**
6	Silent commands now answer	Without pairing: `RTL_LORA_AUTO ON`, `RTL_WEB ON`.	`RTL_LORA_AUTO_ERROR auth_required`, `RTL_WEB_ERROR auth_required`.	**PASS**	Both refused with the exact expected strings.
7	Live screenshot	`python tools/screenshot_tab5.py` (snapshot mode) on a live screen.	`UI_SNAPSHOT_DONE` and a BMP on the PC.	**PASS**	`screenshot_tab5.py --live now` saved a 1280x720 BMP showing a live FM spectrum.

9. SD card and data packs

#	Test	How	Expected	P / F	Notes
1	Free space is real	Settings - Storage.	SD free is **less than** the card size (it used to equal it).	[ ] / [ ]	
2	Install a pack	Data & Maps - CHECK FOR UPDATES - install FAA AVIATION.	Installs; message names any failure.	[ ] / [ ]	
3	Remove frees space	Remove that pack, then `RTL_STOP` and `SD_LIST` (auth).	`SD_LIST_DONE count=N`, with no `.bak` or `.ver` left behind for that pack. `SD_LIST` **takes no path** - the dispatcher matches the verb exactly, so `SD_LIST /orcsdr/data` answers `unknown_command`. It also answers `SD_LIST_ERROR radio_busy` while reception is running, hence the `RTL_STOP` first.	**PART**	The listing itself works: `SD_LIST_DONE count=13` after `RTL_STOP`. **Installing and removing a pack is still to do** (row 2).
4	Copy scripts on PowerShell 5.1	`powershell -File tools\copy_to_tab5_sd.ps1 ...` (Windows PowerShell, not pwsh).	Copies (pairs first). Before, these failed on 5.1 and never authenticated.	[ ] / [ ]	
5	Maps	Check the map list.	Older `lane_county_map.idx` shows UNSUPPORTED FORMAT; FILE PRESERVED (expected).	[ ] / [ ]	

When you are done

Send me the serial log and this sheet's FAIL rows, plus the partition headroom. The green rows are already settled, so what is actually outstanding is: the touch rows in 3, 5 and 6, a signal SNR reading with a handheld (6.3) and the threshold that follows from it (6.4), a scan against a real transmitter (6.5-6.8, 6.11), a SAME burst (7.2), 15 minutes of ADS-B (8.4), and the data-pack rows (9.1, 9.2, 9.4, 9.5).

The **Untested build** warning at the top of `README.md` is now out of date: the build compiles with 0 errors, boots, and passes 71 of 71 self-checks on hardware. It should go.